Sungsoo Ahn

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Education

Integrated M.S.–Ph.D. Program in Electrical Engineering, KAIST, Daejeon, South Korea, Advisor: Jinwoo Shin, Mar. 2015 – Mar. 2021.

B.S. in Electrical Engineering, KAIST, Daejeon, South Korea, Mar. 2011 – Mar. 2015.

Employment

Assistant Professor, KAIST, Graduate School of AI, Daejeon, South Korea, Jan. 2025 – Present.

Assistant Professor, POSTECH, Pohang, South Korea, Graduate School of AI and Department of Computer Science and Engineering, Dec. 2021 – Jan. 2025.

Postdoctoral Research Associate, Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, United Arab Emirates, Supervisors: Le Song and Eric Xing, Mar. 2021 – Nov. 2021.

Research Intern, Amazon Cambridge Development Center, Cambridge, England, Supervisor: Zhenwen Dai, Jun. 2018 – Aug. 2018.

Visiting Student, University of Cambridge, Cambridge, England, Host: Adrian Weller, Mar. 2018 – May 2018.

Research Intern, Los Alamos National Laboratory, Los Alamos, New Mexico, United States, Host: Michael Chertkov, Jun. 2017 – Aug. 2017.

Research Intern, Los Alamos National Laboratory, Los Alamos, New Mexico, United States, Host: Michael Chertkov, Jun. 2016 – Aug. 2016.

Software Developer Intern, Penta Security Systems, Seoul, South Korea, Jan. 2014 – Feb. 2014.

Publications

*Equal contribution. †Equal corresponding authors.

Conference Papers

1. Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, Chanyoung Park, and **Sungsoo Ahn**, Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction, in *International Conference on Machine Learning (ICML)*, 2026. [arXiv]

2. Kiyoung Seong, **Sungsoo Ahn**, Sehui Han, and Changyoung Park, Multimodal Crystal Flow: Any-to-Any Modality Generation for Unified Crystal Modeling, in *International Conference on Machine Learning (ICML)*, 2026. [arXiv]
3. Dongyeop Woo, Marta Skreta, Seonghyun Park, Kirill Neklyudov†, and **Sungsoo Ahn**†, Riemannian MeanFlow, in *International Conference on Machine Learning (ICML)*, 2026. [arXiv]
4. Seongsu Kim, Chanhui Lee, Yoonho Kim, Seongjun Yun, Honghui Kim, Nayoung Kim, Changyoung Park, Sehui Han, Sungbin Lim†, and **Sungsoo Ahn**†, Machine Learning Hamiltonians are Accurate Energy-Force Predictors, in *International Conference on Machine Learning (ICML)*, 2026. [arXiv]
5. Minkyu Kim, Nayoung Kim, Honghui Kim, and **Sungsoo Ahn**, CatFlow: Co-generation of Slab-Adsorbate Systems via Flow Matching, in *International Conference on Machine Learning (ICML)*, 2026. [arXiv]
6. Taewon Kim, Jihwan Shin, Hyomin Kim, Youngmok Jung, Jonghoon Lee, Won-Chul Lee, **Sungsoo Ahn**†, and Insu Han†, DNACHUNKER: Learnable Tokenization for DNA Language Models, in *International Conference on Machine Learning (ICML)*, 2026. [arXiv]
7. Minsu Kim, Jean-Pierre R. Falet, Oliver Ethan Richardson, Xiaoyin Chen, Moksh Jain, Sungjin Ahn, **Sungsoo Ahn**, and Yoshua Bengio, Latent Veracity Inference for Identifying Errors in Stepwise Reasoning, in *International Conference on Learning Representations (ICLR)*, 2026. [arXiv]
8. Hyunjin Seo*, Taewon Kim*, Sihyun Yu, and **Sungsoo Ahn**, Learning Flexible Forward Trajectories for Masked Molecular Diffusion, in *International Conference on Learning Representations (ICLR)*, 2026. [arXiv]
9. Seonghyun Park, Kiyoung Seong, Soojung Yang, Rafael Gómez-Bombarelli, and **Sungsoo Ahn**, Learning Collective Variables from BioEmu with Time-Lagged Generation, in *International Conference on Learning Representations (ICLR)*, 2026. [arXiv]
10. Dongyeop Woo, Minsu Kim, Minkyu Kim, Kiyoung Seong, and **Sungsoo Ahn**, Energy-based Generator Matching: A Neural Sampler for General State Space, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2025. [arXiv]
11. Minkyu Kim, Kiyoung Seong, Dongyeop Woo, **Sungsoo Ahn**, and Minsu Kim, On Scalable and Efficient Training of Diffusion Samplers, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2025. [arXiv]
12. Nayoung Kim, Seongsu Kim, and **Sungsoo Ahn**, Flexible MOF Generation with Torsion-Aware Flow Matching, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2025. [arXiv]
13. Seongsu Kim, Nayoung Kim, Dongwoo Kim, and **Sungsoo Ahn**, High-order Equivariant Flow Matching for Density Functional Theory Hamiltonian Prediction, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2025, spotlight presentation. [arXiv]
14. Hyosoon Jang, Yunhui Jang, Sungjae Lee, Jungseul Ok, and **Sungsoo Ahn**, Self-Training

Large Language Models with Confident Reasoning, in *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025. [arXiv]

15. Hyomin Kim, Yunhui Jang, and **Sungsoo Ahn**, MT-Mol: Multi Agent System with Tool-based Reasoning for Molecular Optimization, in *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025. [arXiv]

16. Yunhui Jang, Jaehyung Kim, and **Sungsoo Ahn**, Improving Chemical Understanding of LLMs via SMILES Parsing, in *Empirical Methods in Natural Language Processing (EMNLP)*, 2025. [arXiv]

17. Federico Berto, Chuanbo Hua, Junyoung Park, Laurin Luttmann, Yining Ma, Fanchen Bu, Jiarui Wang, Haoran Ye, Minsu Kim, Sanghyeok Choi, Nayeli Gast Zepeda, André Hottung, Jianan Zhou, Jieyi Bi, Yu Hu, Fei Liu, Hyeonah Kim, Jiwoo Son, Haeyeon Kim, Davide Angioni, Wouter Kool, Zhiguang Cao, Qingfu Zhang, Joungho Kim, Jie Zhang, Kijung Shin, Cathy Wu, **Sungsoo Ahn**, Guojie Song, Changhyun Kwon, Kevin Tierney, Lin Xie, and Jinkyoo Park, RL4CO: an Extensive Reinforcement Learning for Combinatorial Optimization Benchmark, in *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), Datasets and Benchmarks Track*, 2025. [arXiv]

18. Dongyoon Hahm, Woogyeol Jin, June Suk Choi, **Sungsoo Ahn**, and Kimin Lee, Enhancing LLM Agent Safety via Causal Influence Prompting, in *Findings of the Association for Computational Linguistics: ACL 2025*, 2025. [arXiv]

19. Yunhui Jang, Jaehyung Kim, and **Sungsoo Ahn**, Structural Reasoning Improves Molecular Understanding of LLM, in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. [arXiv]

20. Seonghwan Seo, Minsu Kim, Tony Shen, Martin Ester, Jinkyoo Park, **Sungsoo Ahn**, and Woo Youn Kim, Generative Flows on Synthetic Pathway for Drug Design, in *International Conference on Learning Representations (ICLR)*, 2025. [arXiv]

21. Nayoung Kim, Seongsu Kim, Minsu Kim, Jinkyoo Park, and **Sungsoo Ahn**, MOFFlow: Flow Matching for Structure Prediction of Metal-Organic Frameworks, in *International Conference on Learning Representations (ICLR)*, 2025. [arXiv]

22. Kiyoung Seong, Seonghyun Park, Seonghwan Kim, Woo Youn Kim, and **Sungsoo Ahn**, Transition Path Sampling with Improved Off-Policy Training of Diffusion Path Samplers, in *International Conference on Learning Representations (ICLR)*, 2025. [arXiv]

23. Taewon Kim*, Hyunjin Seo*, **Sungsoo Ahn**, and Eunho Yang, REBIND: Enhancing Ground-state Molecular Conformation Prediction via Force-Based Graph Rewiring, in *International Conference on Learning Representations (ICLR)*, 2025. [arXiv]

24. Minsu Kim*, Sanghyeok Choi*, Taeyoung Yun, Emmanuel Bengio, Leo Feng, Jarrid Rector-Brooks, **Sungsoo Ahn**, Jinkyoo Park, Nikolay Malkin, and Yoshua Bengio, Adaptive Teachers for Amortized Samplers, in *International Conference on Learning Representations (ICLR)*, 2025. [paper]

25. Hyosoon Jang, Yunhui Jang, Minsu Kim, Jinkyoo Park, and **Sungsoo Ahn**, Pessimistic

Backward Policy for GFlowNets, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2024. [paper]

26. Jaeseung Heo, Seungbeom Lee, **Sungsoo Ahn**, and Dongwoo Kim, EPIC: Graph Augmentation with Edit Path Interpolation via Learnable Cost, in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2024. [paper]

27. Hyomin Kim, Yunhui Jang, Jaeho Lee, and **Sungsoo Ahn**, Hybrid Neural Representations for Spherical Data, in *International Conference on Machine Learning (ICML)*, 2024. [paper]

28. Seongsu Kim and **Sungsoo Ahn**, Gaussian Plane-Wave Neural Operator for Electron Density Estimation, in *International Conference on Machine Learning (ICML)*, 2024. [paper]

29. Nayeong Kim, Juwon Kang, **Sungsoo Ahn**, Jungseul Ok, and Suha Kwak, Improving Robustness to Multiple Spurious Correlations by Multi-Objective Optimization, in *International Conference on Machine Learning (ICML)*, 2024. [paper]

30. Hyeonah Kim, Minsu Kim, **Sungsoo Ahn**, and Jinkyoo Park, Symmetric Replay Training: Enhancing Sample Efficiency in Deep Reinforcement Learning for Combinatorial Optimization, in *International Conference on Machine Learning (ICML)*, 2024. [arXiv]

31. Fanchen Bu, Hyeonsoo Jo, Soo Yong Lee, **Sungsoo Ahn**, and Kijung Shin, Tackling Prevalent Conditions in Unsupervised Combinatorial Optimization: Cardinality, Minimum, Covering, and More, in *International Conference on Machine Learning (ICML)*, 2024. [paper]

32. Youngsik Yoon, Gangbok Lee, **Sungsoo Ahn**, and Jungseul Ok, Breadth-First Exploration on Adaptive Grid for Reinforcement Learning, in *International Conference on Machine Learning (ICML)*, 2024. [paper]

33. Hyosoon Jang, Minsu Kim, and **Sungsoo Ahn**, Learning Energy Decompositions for Partial Inference in GFlowNets, in *International Conference on Learning Representations (ICLR)*, 2024, oral presentation (86 of 7,304 submissions, 1.2%). [paper]

34. Yunhui Jang, Seul Lee, and **Sungsoo Ahn**, A Simple and Scalable Representation for Graph Generation, in *International Conference on Learning Representations (ICLR)*, 2024. [paper]

35. Yunhui Jang, Dongwoo Kim, and **Sungsoo Ahn**, Graph Generation with K^2-trees, in *International Conference on Learning Representations (ICLR)*, 2024. [paper]

36. Minsu Kim, Taeyoung Yun, Emmanuel Bengio, Dinghuai Zhang, Yoshua Bengio, **Sungsoo Ahn**, and Jinkyoo Park, Local Search GFlowNets, in *International Conference on Learning Representations (ICLR)*, 2024, spotlight presentation (366 of 7,304 submissions, 5.0%). [paper]

37. Hyuna Cho, Minjae Jeong, Sooyeon Jeon, **Sungsoo Ahn**, and Won Hwa Kim, Multi-resolution Spectral Coherence for Graph Generation with Score-based Diffusion, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2023. [paper]

38. Hyosoon Jang, Seonghyun Park, Sangwoo Mo, and **Sungsoo Ahn**, Diffusion Probabilistic Models for Structured Node Classification, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2023. [paper]

39. Minsu Kim, Federico Berto, **Sungsoo Ahn**, and Jinkyoo Park, Bootstrapped Training of Score-Conditioned Generator for Offline Design of Biological Sequences, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2023. [paper]
40. Sungbin Shin, Yohan Jo, **Sungsoo Ahn**, and Namhoon Lee, A Closer Look at the Intervention Procedure of Concept Bottleneck Models, in *International Conference on Machine Learning (ICML)*, 2023. [paper]
41. Junsu Kim, Younggyo Seo, **Sungsoo Ahn**, Kyunghwan Son, and Jinwoo Shin, Imitating Graph-Based Planning with Goal-Conditioned Policies, in *International Conference on Learning Representations (ICLR)*, 2023. [paper]
42. Nayeong Kim, Sehyun Hwang, **Sungsoo Ahn**, Jaesik Park, and Suha Kwak, Learning Debiased Classifier with Biased Committee, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2022. [paper]
43. Kyunghwan Son, Junsu Kim, **Sungsoo Ahn**, Roben Delos Reyes, Yung Yi, and Jinwoo Shin, Disentangling Sources of Risk for Distributional Multi-Agent Reinforcement Learning, in *International Conference on Machine Learning (ICML)*, 2022. [paper]
44. Jaehyung Kim, Dongyeop Kang, **Sungsoo Ahn**, and Jinwoo Shin, What Makes Better Augmentation Strategies? Augment Difficult but Not Too Different, in *International Conference on Learning Representations (ICLR)*, 2022. [paper]
45. **Sungsoo Ahn**, Binghong Chen, Tianzhe Wang, and Le Song, Spanning Tree-based Graph Generation for Molecules, in *International Conference on Learning Representations (ICLR)*, 2022, spotlight presentation (174 of 3,422 submissions, 5.1%). [paper]
46. Sihyun Yu, **Sungsoo Ahn**, Le Song, and Jinwoo Shin, RoMA: Robust Model Adaptation for Offline Model-based Optimization, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2021. [arXiv]
47. Hankook Lee, **Sungsoo Ahn**, Seung-Woo Seo, You Young Song, Eunho Yang, Sung-Ju Hwang, and Jinwoo Shin, RetCL: A Selection-based Approach for Retrosynthesis via Contrastive Learning, in *International Joint Conference on Artificial Intelligence (IJCAI)*, 2021. [paper]
48. Junsu Kim, **Sungsoo Ahn**, Hankook Lee, and Jinwoo Shin, Self-Improved Retrosynthetic Planning, in *International Conference on Machine Learning (ICML)*, 2021. [paper]
49. Jaeho Lee, Sejun Park, Sangwoo Mo, **Sungsoo Ahn**, and Jinwoo Shin, Layer-adaptive sparsity for the Magnitude-based Pruning, in *International Conference on Learning Representations (ICLR)*, 2021. [paper]
50. Junhyun Nam, Hyuntak Cha, **Sungsoo Ahn**, Jaeho Lee, and Jinwoo Shin, Learning from Failure: Training Debiased Classifier from Biased Classifier, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2020. [paper]
51. **Sungsoo Ahn**, Junsu Kim, Hankook Lee, and Jinwoo Shin, Guiding Deep Molecular Optimization with Genetic Exploration, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2020. [paper]

52. **Sungsoo Ahn**, Younggyo Seo, and Jinwoo Shin, Learning What to Defer for Maximum Independent Sets, in *International Conference on Machine Learning (ICML)*, 2020. [paper]

53. **Sungsoo Ahn**, Shell Xu Hu, Andreas Damianou, Neil D. Lawrence, and Zhenwen Dai, Variational Information Distillation for Knowledge Transfer, in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. [paper]

54. **Sungsoo Ahn**, Michael Chertkov, Adrian Weller, and Jinwoo Shin, Bucket Renormalization for Approximate Inference, in *International Conference on Machine Learning (ICML)*, 2018. [paper]

55. **Sungsoo Ahn**, Michael Chertkov, Jinwoo Shin, and Adrian Weller, Gauged Mini-Bucket Elimination for Approximate Inference, in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2018. [paper]

56. **Sungsoo Ahn**, Michael Chertkov, and Jinwoo Shin, Gauging Variational Inference, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2017. [paper]

57. **Sungsoo Ahn**, Michael Chertkov, and Jinwoo Shin, Synthesis of MCMC and Belief Propagation, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2016, oral presentation (46 of 2,500 submissions, 1.8%). [paper]

58. **Sungsoo Ahn**, Sejun Park, Michael Chertkov, and Jinwoo Shin, Minimum Weight Perfect Matching via Blossom Belief Propagation, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2015, spotlight presentation (82 of 1,838 submissions, 4.5%). [paper]

Journal Articles

1. Nayoung Kim, Minsu Kim, **Sungsoo Ahn**, and Jinkyoo Park, Decoupled Sequence and Structure Generation for Realistic Antibody Design, *Transactions of Machine Learning Research (TMLR)*, 2024. [paper]

2. Seonghyun Park*, Narae Ryu*, Gahee Kim, Dongyeop Woo, Se-Young Yun†, and **Sungsoo Ahn**†, Non-backtracking Graph Neural Networks, *Transactions of Machine Learning Research (TMLR)*, 2024. [paper]

3. Seojin Kim, Jaehyun Nam, Junsu Kim, Hankook Lee, **Sungsoo Ahn**, and Jinwoo Shin, Holistic Molecular Representation Learning via Multi-view Fragmentation, *Transactions of Machine Learning Research (TMLR)*, 2024. [paper]

4. **Sungsoo Ahn**, Michael Chertkov, Adrian Weller, and Jinwoo Shin, Bucket Renormalization for Approximate Inference, *Journal of Statistical Mechanics: Theory and Experiment*, 2019(12), 124015, 2019. [paper]

5. **Sungsoo Ahn**, Michael Chertkov, and Jinwoo Shin, Gauging Variational Inference, *Journal of Statistical Mechanics: Theory and Experiment*, 2019(12), 124022, 2019. [paper]

6. **Sungsoo Ahn**, Michael Chertkov, Andrew E. Gelfand, Sejun Park, and Jinwoo Shin, Maximum Weight Matching using Odd-sized Cycles: Max-Product Belief Propagation and Half-Integrality, *IEEE Transactions on Information Theory*, 64(3), 1471–1480, 2018. [paper]

Preprints

1. Kiyoung Seong, Nayoung Kim, and **Sungsoo Ahn**, Discovering Crystal Structure Prediction Algorithms with an AI Co-Scientist, *arXiv*, 2026. [arXiv]

2. Yoonho Kim, Seongsu Kim, **Sungsoo Ahn**, and Honghui Kim, MADField: Multi-fidelity Amortized Density Field for Adsorption in Nanoporous Materials, *arXiv*, 2026. [arXiv]

3. Taewon Kim, Hyosoon Jang, Hyunjin Seo, Seonghwan Seo, Hyeongwoo Kim, Wonho Zhung, Mingyeong Shin, Wooyoun Kim, and **Sungsoo Ahn**, Atom-level Protein Representation Learning Improves Protein Structure Prediction, *arXiv*, 2026. [arXiv]

4. Hyunjin Seo, Hongjoon Ahn, Jimin Park, Sungjun Han, Gyubok Lee, Soojung Yang, Joseph S Brown, Leo Chen, Gina El Nesr, Feyisayo Eweje, Sarah Gurev, Hyejin Lee, Cheng-Hao Liu, Junlang Liu, Zhihui Qi, Gyu Rie Lee, **Sungsoo Ahn**, Jamin Shin, and Sangwon Jung, VibeProteinBench: An Evaluation Benchmark for Language-interfaced Vibe Protein Design, *arXiv*, 2026. [arXiv]

5. Hyomin Kim, Sang-Yeon Hwang, Jaechang Lim, Yinhua Piao, Yunhak Oh, Woo Youn Kim, Chanyoung Park, **Sungsoo Ahn**, and Junhyeok Jeon, Progressive Multi-Agent Reasoning for Biological Perturbation Prediction, *arXiv*, 2026. [arXiv]

6. Nayoung Kim, Honghui Kim, Sihyun Yu, Minkyu Kim, Seongsu Kim, and **Sungsoo Ahn**, AtomMOF: All-Atom Flow Matching for MOF-Adsorbate Structure Prediction, *arXiv*, 2026. [arXiv]

7. Yunhui Jang, Seonghyun Park, Jaehyung Kim, and **Sungsoo Ahn**, INDIBATOR: Diverse and Fact-Grounded Individuality for Multi-Agent Debate in Molecular Discovery, *arXiv*, 2026. [arXiv]

8. Hyosoon Jang, Hyunjin Seo, Honghui Kim, Seonghyun Park, Taewon Kim, Yunhui Jang, and **Sungsoo Ahn**, A Systematic Evaluation of Co-folding Model Representations for Small-Molecule Learning, *arXiv*, 2026. [arXiv]

9. Hyosoon Jang, Yunhui Jang, Jaehyung Kim, and **Sungsoo Ahn**, Can LLMs Generate Diverse Molecules? Towards Alignment with Structural Diversity, *arXiv*, 2024. [arXiv]

10. Dongyeop Woo and **Sungsoo Ahn**, Iterated Energy-based Flow Matching for Sampling from Boltzmann Densities, *arXiv*, 2024. [arXiv]

Service

1. *Track Chair (AI Scientist / Agent Systems)*, AI for Science Conference (AI4Sci Korea), Seoul, South Korea, 2026.

2. *Host*, Korea AI4Science Gathering @ ICML, Seoul, South Korea, 2026.

3. *Local Chair*, Symposium on Probabilistic Machine Learning (ProbML), Seoul, South Korea, 2026.

4. *Organizer*, AI for Science: AI Scientists—Tools, Co-authors, or Founders?, International Conference on Machine Learning (ICML), 2026.

5. *Mentor*, AI Co-Research & Education for Innovative Drug (AI-CRED) Institute, InnoCORE Program, Project N10250153, 2025 – Present.
6. *Board of Directors*, Korean Society of AI Frontier (KSAIF), 2025 – Present.
7. *Area Chair*, International Conference on Machine Learning (ICML), 2022 – 2026.
8. *Area Chair*, International Conference on Learning Representations (ICLR), 2024 – 2026.
9. *Area Chair*, Conference on Neural Information Processing Systems (NeurIPS), 2022 – 2026.
10. *Area Chair*, International Conference on Artificial Intelligence and Statistics (AISTATS), 2022 – 2024.
11. *Area Chair*, AAAI Conference on Artificial Intelligence (AAAI), 2024.
12. *Action Editor*, Transactions of Machine Learning Research (TMLR), 2024 – Present.
13. *Workflow Chair*, International Conference on Machine Learning (ICML), 2022.
14. *Reviewer*, NeurIPS, 2018 – 2021.
15. *Reviewer*, ICML, 2019 – 2021.
16. *Reviewer*, ICLR, 2019 – 2024.
17. *Reviewer*, AISTATS, 2019 – 2021.

Talks

1. *비정형 데이터 기반 딥러닝의 과학기술 활용 및 혁신 사례*, KAIST Chief AI Officer Program, Daejeon, South Korea, Nov. 2026.
2. *From Atomistic Generative Models to Amortized Free-Energy Estimation*, Generative Models and Uncertainty Quantification (GenU), Copenhagen, Denmark, Sep. 2026.
3. *AI for Molecular and Materials Discovery*, KAIST College of AI Colloquium, Daejeon, South Korea, Sep. 2026.
4. *From Atomistic Generative Models to Amortized Free-Energy Estimation*, KAIST–Mila Prefrontal AI Workshop, Aug. 2026.
5. *K-Fold: Fast Biomolecular Complex Prediction without Evolutionary Signals*, Korean Society of Artificial Intelligence in Medicine (KoSAIM), Jul. 2026.
6. *Machine Learning Hamiltonians*, 22nd KIAS Workshop on Electronic Structure Calculations, Seoul, South Korea, Jul. 2026.
7. *Generative Approaches to Protein Conformational Landscapes*, 2026 Korean Biomolecular Science Joint Conference, South Korea, Jun. 2026.
8. *Generative Approaches to Protein Conformational Landscapes*, Korea Institute for Advanced Study (KIAS), Seoul, South Korea, Jun. 2026.

9. *비정형 데이터 기반 딥러닝의 과학기술 활용 및 혁신 사례*, KAIST Chief AI Officer Program, Daejeon, South Korea, May 2026.
10. *AI Methods for Generating Scientific Hypotheses*, Samsung Advanced Institute of Technology Scientific AI Workshop, Suwon, South Korea, May 2026.
11. *Generative Models for Molecular Discovery*, Korea Institute of Science and Technology Information (KISTI), Daejeon, South Korea, Mar. 2026.
12. *Foundation Models for Material Discovery*, Korea Institute of Materials Science (KIMS), Changwon, South Korea, Jul. 2025.
13. *Generative Models for Molecular Discovery*, Seoul National University, Seoul, South Korea, Jul. 2025.
14. *A Generative Model for Metal-Organic Frameworks*, KAIST–Mila Annual Workshop, Montreal, Canada, Jul. 2025.
15. *Machine Learning for Transition Path Sampling*, The Korea in Silico BioDesign and Discovery Society (KIDDS), Daejeon, South Korea, Jul. 2025.
16. *Generative Models for Molecular Discovery*, LG AI Research, Seoul, South Korea, May 2025.
17. *Optimization Methods for Materials Science*, POSCO Holdings, Seoul, South Korea, Dec. 2024.
18. *Structure- and Energy-Based Machine Learning for Molecules*, KAIST–Mila Prefrontal AI Workshop, Montreal, Canada, Dec. 2024.
19. *Amortizing Intractable Inference for Molecular Discovery*, Physics-Informed Machine Learning Workshop, New Mexico, United States, Nov. 2024.
20. *Opportunities and Challenges in Deep Graph Generative Models*, AI Korea, Seoul, South Korea, Nov. 2023.
21. *Machine Learning for Drug Discovery*, KAIST, Daejeon, South Korea, Oct. 2023.
22. *Graph Structured Prediction and Graph Generation with Deep Learning*, KAIST, Daejeon, South Korea, Jun. 2023.
23. *Deep Neural Networks for Graph Optimization*, University of Arizona, Tucson, Arizona, United States, Nov. 2022.
24. *Geometric Deep Learning for Drug Discovery*, Korean Artificial Intelligence Association (KAIA), Jeju, South Korea, Aug. 2022.
25. *Machine Learning for Drug Discovery*, Centre for Frontier AI Research, Singapore, Jun. 2022.
26. *Elimination Techniques in Probabilistic Graphical Models*, Skolkovo Institute of Science and Technology (Skoltech), Moscow, Russia, Jun. 2021.
27. *Guiding Deep Molecular Optimization with Genetic Exploration*, Seminar at Samsung Advanced Institute of Technology, Suwon, South Korea, Dec. 2020.

28. *Scaling Deep Reinforcement Learning to Large Combinatorial Problems*, AI & CSE Seminar, POSTECH, Pohang, South Korea, Feb. 2020.
29. *Bucket Renormalization for Approximate Inference*, Robotics Seminar, University of Oxford, Oxford, England, Jul. 2018.
30. *Mini-Bucket Renormalization*, Physics-Informed Machine Learning Workshop, Santa Fe, New Mexico, United States, Feb. 2018.
31. *Gauge Transformation of Graphical Models*, CNLS Student Seminar, Los Alamos National Laboratory, Los Alamos, New Mexico, United States, Jul. 2017.
32. *Optimizing Gauge Transformation for Inference in Graphical Models*, Banff Workshop on Optimization and Inference for Physical Flows on Networks, Banff, Alberta, Canada, Feb. 2017.
33. *Synthesis of MCMC and Belief Propagation*, CNLS Student Seminar, Los Alamos National Laboratory, Los Alamos, New Mexico, United States, Jul. 2016.
34. *Minimum Weight Perfect Matching via Blossom Belief Propagation*, Discrete Math Seminar, KAIST, Daejeon, South Korea, Nov. 2015.

Courses

1. *Proteins and AI (CoE499)*, KAIST, Spring 2026.
2. *Machine Learning for Molecules (AI899)*, KAIST, Fall 2025.
3. *Geometric Deep Learning (AI810)*, KAIST, Spring 2025.
4. *Introduction to Artificial Intelligence (CSED105)*, POSTECH, Fall 2024.
5. *Machine Learning for Graphs (CSED/AIGS703I)*, POSTECH, Spring 2024.
6. *Introduction to Artificial Intelligence (CSED105)*, POSTECH, Fall 2023.
7. *Probabilistic Graphical Models (CSED/AIGS524)*, POSTECH, Spring 2023.
8. *Introduction to Machine Learning (CSED490B)*, POSTECH, Fall 2022.
9. *Probabilistic Graphical Models (CSED/AIGS524)*, POSTECH, Spring 2022.

Awards

1. *Outstanding Teaching Award*, KAIST, 2026.
2. *Q-Day Faculty Special Award, Research Division*, KAIST, 2025.

Grants

Research

1. *Developing a Highly Multimodal (Drug, Protein, Gene, Cell Imaging, Literature) Foundation Model for ADMET Property Prediction*, Principal Investigator, K-MELLODDY, Korea Health Industry Development Institute, Jul. 2025 – Dec. 2027. KRW 750M total project funding, Project RS-2025-21063030.
2. *AI Star Fellowship: Virtual-Cell Foundation Models and Cell-Ontology Graphs*, Project 3 Principal Investigator, Institute for Information & Communications Technology Planning & Evaluation, Apr. 2025 – Dec. 2030. KRW 11.5B total project funding, Project RS-2025-02304967.

3. *Geometric Deep Learning for Analysis of Unstructured Neuroimaging Data*, Co-Investigator, National Research Foundation of Korea, Global Basic Research Laboratory Program, Jun. 2025 – May 2028. KRW 1.5B total project funding, Project RS-2025-02216257.
4. *KAIST–Mila Prefrontal AI Research Center*, Co-Principal Investigator (2024), Co-Investigator (2025 – Present), National Research Foundation of Korea, Jul. 2024 – Dec. 2028. KRW 2.0B total project funding, Project RS-2024-00436165.
5. *Three-Dimensional Graph Neural Networks for Long-Range Interactions in Crystals*, Principal Investigator, POSCO Holdings, 2023 – 2024. KRW 50M total project funding.
6. *Deep Learning for Molecules to Accelerate Drug Discovery*, Principal Investigator, National Research Foundation of Korea, Outstanding Young Researcher Program, 2022 – 2027. KRW 600M total direct costs, Project RS-2022-NR072184.
7. *AI-Based Recognition of Three-Dimensional Molecular Structures*, Principal Investigator, POSCO Holdings, 2022 – 2023. KRW 49.3M total project funding.

Computing

1. *Towards a Unified Generative Model for Molecular Interactions in Materials*, NVIDIA Academic Grant Program, Jul. 2026 – Dec. 2026. 8 NVIDIA H100 GPUs.
2. *Development of a Materials Generative Foundation Model with Molecular-Materials Interactions*, Advanced GPU Utilization Support Program, Ministry of Science and ICT, Mar. 2026 – Dec. 2026. 56 NVIDIA H200 GPUs.
3. *An Architecture for DFT-Level General-Purpose Atomistic Simulation Based on Simultaneous Hamiltonian/Density Prediction*, KISTI Supercomputing Resources, National Supercomputing Center, Jan. 2026 – Dec. 2026. 36 Nurion KNL and 10 Neuron allocations, Project KSC-2025-CRE-0602.
4. *Generative Materials Foundation Models for Adsorbate Complex Design*, Advanced GPU Utilization Support Program, Beta Service, Ministry of Science and ICT, Jan. 2026 – Feb. 2026. 32 NVIDIA B200 GPUs.
5. *Integrated Representation Learning for Biomolecular Foundation Models*, AI Specialized Foundation Model Project, National IT Industry Promotion Agency, Nov. 2025 – Oct. 2026. 256 NVIDIA B200 GPUs, Project PJT-25-100009.
6. *Generative Models for Designing Metal-Organic Frameworks for Direct Air Capture*, Regional Infrastructure-Based Computing Resources, AI Industry Convergence Agency, Sep. 2025 – Dec. 2025. 8 NVIDIA H100 GPUs.
7. *Deep Generative Models for Simulation and Design of Metal-Organic Frameworks*, NVIDIA Academic Grant Program, Apr. 2025 – Sep. 2025. 16,000 NVIDIA A100 GPU-hours.
8. *Developing Universal Coarse-Grained Atomistic Foundation Models*, AI Research Computing Support Project, Institute for Information & Communications Technology Planning & Evaluation, 2025. 32 NVIDIA H100 GPUs, Project RS-2025-02653113.
9. *Chemical Language and Logic for Artificial Chemical Intelligence*, KISTI Supercomputing Resources, National Supercomputing Center, Jan. 2025 – Dec. 2025. 10 Neuron allocations, Project KSC-2024-CRE-0535.